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Batch: 15

Experiment-40

Algorithm

1. **Start.**
2. Declare character arrays firstName, middleName, lastName, fullName, copiedName, reverseName, and word.
3. Declare integer variables choice, comparisonResult, vowels, consonants, digits, spaces, and searchResult. Declare length as size_t.
4. Read the student's **first name, middle name, and last name.**
5. Call combineNames() to combine the first, middle, and last names and store the result in fullName.
6. Display the following menu:
 1. Combine names
 2. Find length of full name
 3. Copy full name
 4. Compare first name and last name
 5. Display name in reverse order
 6. Count vowels, consonants, digits, and spaces
 7. Search a word in the full name
 8. Exit
7. Read the user's choice.
8. Use a **switch statement** to perform the selected operation:

Case 1: Combine Names

- Call combineNames().
- Copy firstName into fullName using strcpy().
- Append a space, middleName, another space, and lastName using strcat().
- Display fullName.

Case 2: Find Length

- Call findLength().
- Find the length of fullName using strlen().
- Store the result in length using a pointer.
- Display the length.

Case 3: Copy Full Name

- Call copyName().

- Copy fullName into copiedName using strcpy().
- Display copiedName.

Case 4: Compare First and Last Names

- Call compareNames().
- Compare firstName and lastName using strcmp().
- Store the result in comparisonResult using a pointer.
- If the result is 0, display **names are equal**.
- Otherwise, display **names are different**.

Case 5: Reverse Name Order

- Call reverseNameOrder().
- Copy lastName into reverseName.
- Append middleName and firstName using strcat(), with spaces between them.
- Display the name as **Last Middle First**.

Case 6: Count Characters

- Call countCharacters().
- Initialize vowels, consonants, digits, and spaces to 0 using pointers.
- Traverse each character of fullName.
- If the character is an alphabet:
 - Convert it to lowercase.
 - If it is a, e, i, o, u, increment vowels.
 - Otherwise, increment consonants.
- If it is a digit, increment digits.
- If it is a space, increment spaces.
- Display all four counts.

Case 7: Search Word

- Read the word to be searched.
- Call searchWord().
- Search for the word in fullName using strstr().
- Store the result using a pointer.
- If found, display **Word found**.
- Otherwise, display **Word not found**.

Case 8: Exit

- Display an exit message and terminate the program.

Default:

- Display **Invalid choice**.
9. Repeat Steps 6–8 until the user selects 8 (Exit).
 10. Stop.

Function	Header	Purpose
printf(), scanf()	<stdio.h>	Input/output
strcpy()	<string.h>	Copy string
strcat()	<string.h>	Join strings
strlen()	<string.h>	Find string length
strcmp()	<string.h>	Compare strings
strstr()	<string.h>	Search substring
isalpha()	<ctype.h>	Check alphabet
isdigit()	<ctype.h>	Check digit
isspace()	<ctype.h>	Check space
tolower()	<ctype.h>	Convert character to lowercase

C Program

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

// Function declarations

void combineNames(const char firstName[], const char middleName[],
                 const char lastName[], char fullName[]);

void findLength(const char fullName[], size_t *length);

void copyName(const char source[], char destination[]);

void compareNames(const char firstName[], const char lastName[],
                 int *comparisonResult);

void reverseNameOrder(const char firstName[], const char middleName[],
                    const char lastName[], char reverseName[]);

void countCharacters(const char fullName[], int *vowels,
                   int *consonants, int *digits, int *spaces);
```

```
void searchWord(const char fullName[], const char word[],  
               int *searchResult);
```

```
// 1. Combine names
```

```
void combineNames(const char firstName[], const char middleName[],  
                 const char lastName[], char fullName[])  
{  
    strcpy(fullName, firstName);  
    strcat(fullName, " ");  
    strcat(fullName, middleName);  
    strcat(fullName, " ");  
    strcat(fullName, lastName);  
}
```

```
// 2. Find length
```

```
void findLength(const char fullName[], size_t *length)  
{  
    *length = strlen(fullName);  
}
```

```
// 3. Copy full name
```

```
void copyName(const char source[], char destination[])  
{  
    strcpy(destination, source);  
}
```

// 4. Compare first and last names

```
void compareNames(const char firstName[], const char lastName[],  
                  int *comparisonResult)  
{  
    *comparisonResult = strcmp(firstName, lastName);  
}
```

// 5. Reverse name order

```
void reverseNameOrder(const char firstName[], const char middleName[],  
                      const char lastName[], char reverseName[])  
{  
    strcpy(reverseName, lastName);  
    strcat(reverseName, " ");  
    strcat(reverseName, middleName);  
    strcat(reverseName, " ");  
    strcat(reverseName, firstName);  
}
```

// 6. Count vowels, consonants, digits and spaces

```
void countCharacters(const char fullName[], int *vowels,  
                    int *consonants, int *digits, int *spaces)  
{  
    int i;  
    char ch;  
  
    *vowels = 0;  
    *consonants = 0;  
    *digits = 0;
```

```
*spaces = 0;

for (i = 0; i < strlen(fullName); i++)
{
    ch = tolower(fullName[i]);

    if (isalpha(ch))
    {
        if (ch == 'a' || ch == 'e' || ch == 'i' ||
            ch == 'o' || ch == 'u')
        {
            (*vowels)++;
        }
        else
        {
            (*consonants)++;
        }
    }
    else if (isdigit(ch))
    {
        (*digits)++;
    }
    else if (isspace(ch))
    {
        (*spaces)++;
    }
}
```

```
// 7. Search word

void searchWord(const char fullName[], const char word[],
               int *searchResult)
{
    if (strstr(fullName, word) != NULL)
        *searchResult = 1;
    else
        *searchResult = 0;
}
```

```
// Main function

int main()
{
    char firstName[50], middleName[50], lastName[50];
    char fullName[150], copiedName[150];
    char reverseName[150], word[50];

    int choice;
    int comparisonResult;
    int vowels, consonants, digits, spaces;
    int searchResult;
    size_t length;

    printf("Enter first name: ");
    scanf("%49s", firstName);

    printf("Enter middle name: ");
    scanf("%49s", middleName);
```

```
printf("Enter last name: ");
scanf("%49s", lastName);

// Form full name once
combineNames(firstName, middleName, lastName, fullName);

do
{
    printf("\n----- MENU ----- \n");
    printf("1. Combine names\n");
    printf("2. Find length of full name\n");
    printf("3. Copy full name\n");
    printf("4. Compare first and last names\n");
    printf("5. Display name in reverse order\n");
    printf("6. Count vowels, consonants, digits and spaces\n");
    printf("7. Search a word in full name\n");
    printf("8. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            combineNames(firstName, middleName,
                          lastName, fullName);

            printf("Full Name: %s\n", fullName);
            break;
```

case 2:

```
findLength(fullName, &length);
```

```
printf("Length of full name: %zu\n", length);
```

```
break;
```

case 3:

```
copyName(fullName, copiedName);
```

```
printf("Copied Name: %s\n", copiedName);
```

```
break;
```

case 4:

```
compareNames(firstName, lastName,  
              &comparisonResult);
```

```
if (comparisonResult == 0)
```

```
    printf("First name and last name are equal.\n");
```

```
else
```

```
    printf("First name and last name are different.\n");
```

```
break;
```

case 5:

```
reverseNameOrder(firstName, middleName,  
                  lastName, reverseName);
```

```
printf("Reverse Name Order: %s\n", reverseName);
```

```
break;
```

case 6:

```
countCharacters(fullName, &vowels,  
                &consonants, &digits, &spaces);
```

```
printf("Vowels: %d\n", vowels);  
printf("Consonants: %d\n", consonants);  
printf("Digits: %d\n", digits);  
printf("Spaces: %d\n", spaces);  
break;
```

case 7:

```
printf("Enter word to search: ");  
scanf("%49s", word);  
  
searchWord(fullName, word, &searchResult);  
  
if (searchResult == 1)  
    printf("Word found in the full name.\n");  
else  
    printf("Word not found in the full name.\n");  
  
break;
```

case 8:

```
printf("Exiting program...\n");  
break;
```

default:

```
printf("Invalid choice! Please enter 1 to 8.\n");
```

```
}
```

```

    } while (choice != 8);

    return 0;
}

```

Compilation And Execution

```

(kali㉿kali)-[~]
$ gcc hdoc40.c -o hdoc40

(kali㉿kali)-[~]
$ ./hdoc40
Enter first name: Gaurika
Enter middle name: HAHA
Enter last name: Mittal

—— MENU ——
1. Combine names
2. Find length of full name
3. Copy full name
4. Compare first and last names
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
Enter your choice: 1
Full Name: Gaurika HAHA Mittal

—— MENU ——
1. Combine names
2. Find length of full name
3. Copy full name
4. Compare first and last names
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
Enter your choice: 2
Length of full name: 19

—— MENU ——
1. Combine names
2. Find length of full name
3. Copy full name
4. Compare first and last names
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
Enter your choice: 3
Copied Name: Gaurika HAHA Mittal

—— MENU ——
1. Combine names
2. Find length of full name
3. Copy full name
4. Compare first and last names
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
Enter your choice: 4
First name and last name are different.

—— MENU ——

```

1. Combine names
2. Find length of full name
3. Copy full name
4. Compare first and last names
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit

Enter your choice: 5

Reverse Name Order: Mittal HAHA Gaurika

—— MENU ——

1. Combine names
2. Find length of full name
3. Copy full name
4. Compare first and last names
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit

Enter your choice: 6

Vowels: 8

Consonants: 9

Digits: 0

Spaces: 2

—— MENU ——

1. Combine names
2. Find length of full name

3. Copy full name
4. Compare first and last names
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit

Enter your choice: 7

Enter word to search: HAHA

Word found in the full name.

—— MENU ——

1. Combine names
2. Find length of full name
3. Copy full name
4. Compare first and last names
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit

Enter your choice: 8

Exiting program...